Pose Estimation Using Silhouette and Texture

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1 Introduction

The objective of this work is to estimate the pose of a 3D object by optimizing camera parameters using differentiable rendering. Unlike the previous task, this work incorporates both silhouette-based loss and texture-based (RGB) loss, allowing analysis of their individual contributions to pose estimation accuracy.

2 Objective

The object used in this work is a 3D guitar model in OBJ format.

Figure 1: Object Guitar

Since the model did not include a predefined texture, a synthetic texture was generated. Each triangle (face) of the mesh was assigned a random color using a texture atlas. This approach satisfies the requirement of using a textured object while maintaining simplicity.

To improve computational efficiency, mesh decimation was applied, reducing the number of faces while preserving the overall geometry of the object.

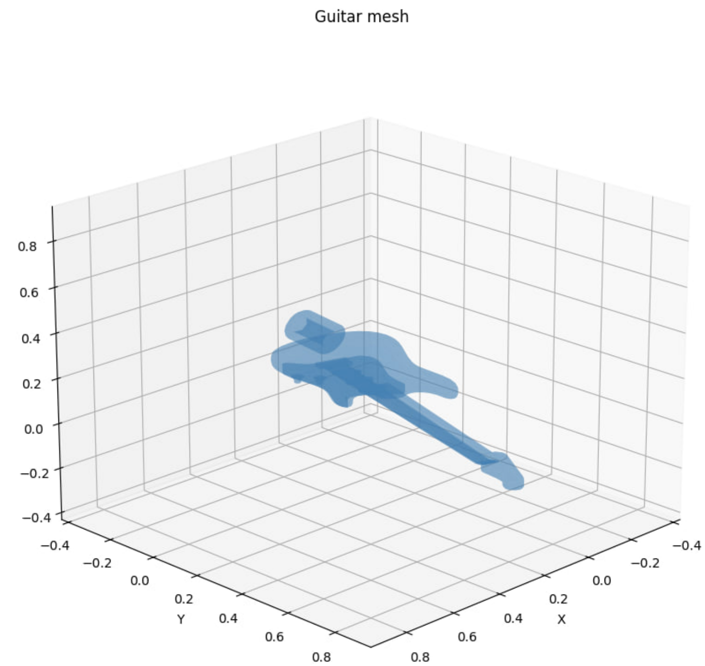

Figure 2: Decimated and Normalized Guitar object

3 Methodology

3.1 Data Preparation

- The 3D model was loaded using PyTorch3D.
- The mesh was simplified using quadric decimation.
- The object was normalized (centered and scaled).
- A synthetic texture was assigned to each triangle.

3.2 Target Image Generation

Target images were generated from a known camera pose: **RGB image**, **Silhouette image**

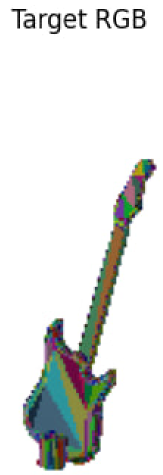

Figure 3: Target RGB

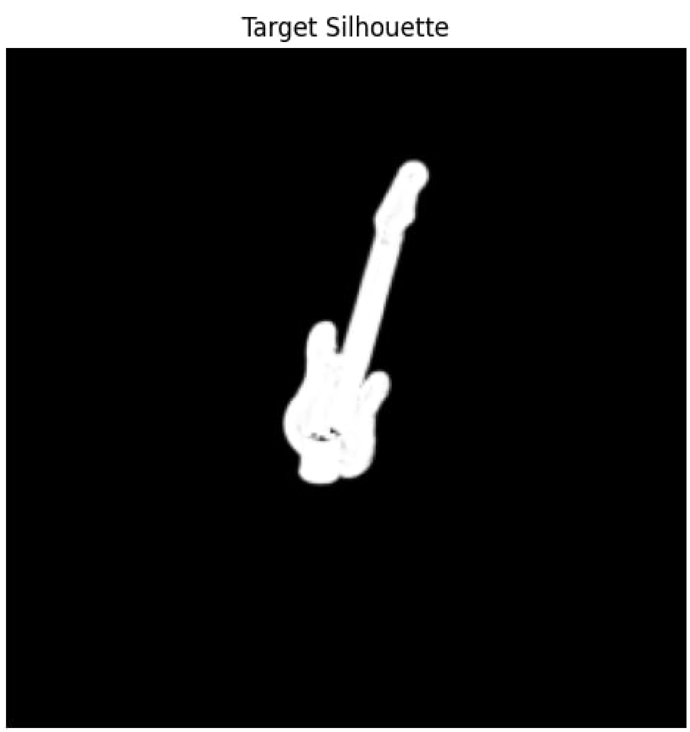

Figure 4: Target silhouette

These images serve as ground truth for optimization.

3.3 Camera Optimization

A differentiable rendering pipeline was used to optimize camera parameters: **Distance**, **Elevation**, **Azimuth**

The optimization minimizes a weighted Mean Squared Error (MSE) loss:

$$\text{Loss} = w_{\text{silh}} \cdot L_{\text{silh}} + w_{\text{tex}} \cdot L_{\text{tex}}$$

where:

- L_{silh} — silhouette loss
- L_{tex} — RGB (texture) loss

3.4 Experiments

Four experiments were conducted with different loss weights:

1. $w_{\text{silh}} = 1,\ w_{\text{tex}} = 0$
2. $w_{\text{silh}} = 0,\ w_{\text{tex}} = 1$
3. $w_{\text{silh}} = 1,\ w_{\text{tex}} = 0.1$
4. $w_{\text{silh}} = 1,\ w_{\text{tex}} = 0.3$

4 Results

4.1 Silhouette Only (1, 0)

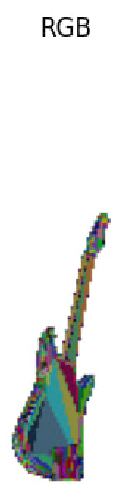

Figure 5: RGB - silh=1, tex=0

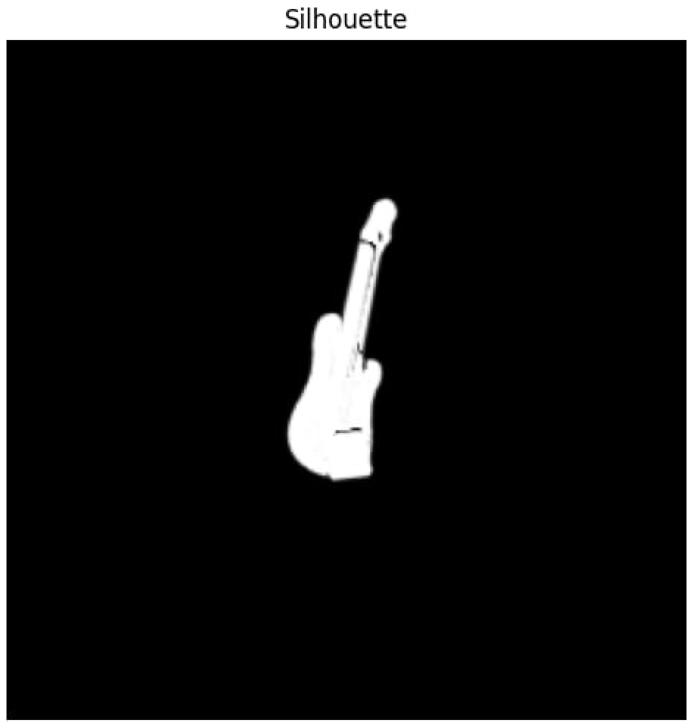

Figure 6: Silhouette - silh=1, tex=0

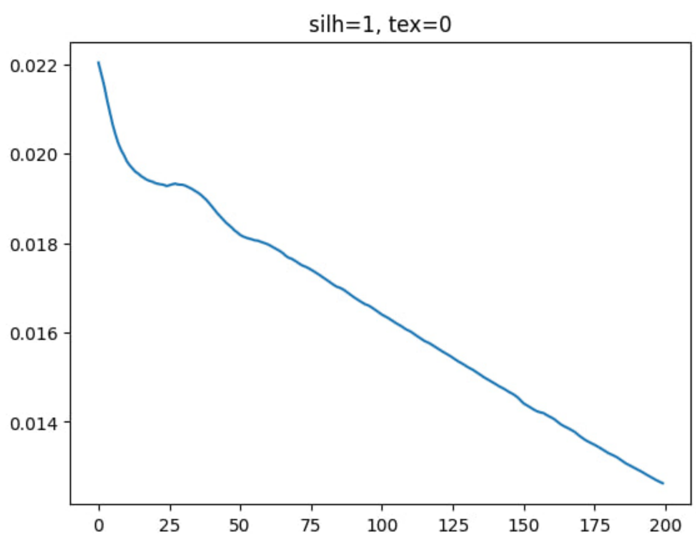

Figure 7: Loss - silh=1, tex=0

- Stable convergence
- Correct object shape and pose
- Smooth loss decrease

Conclusion: Silhouette provides strong global geometric information.

4.2 Texture Only (0, 1)

Figure 8: RGB - silh=0, tex=1

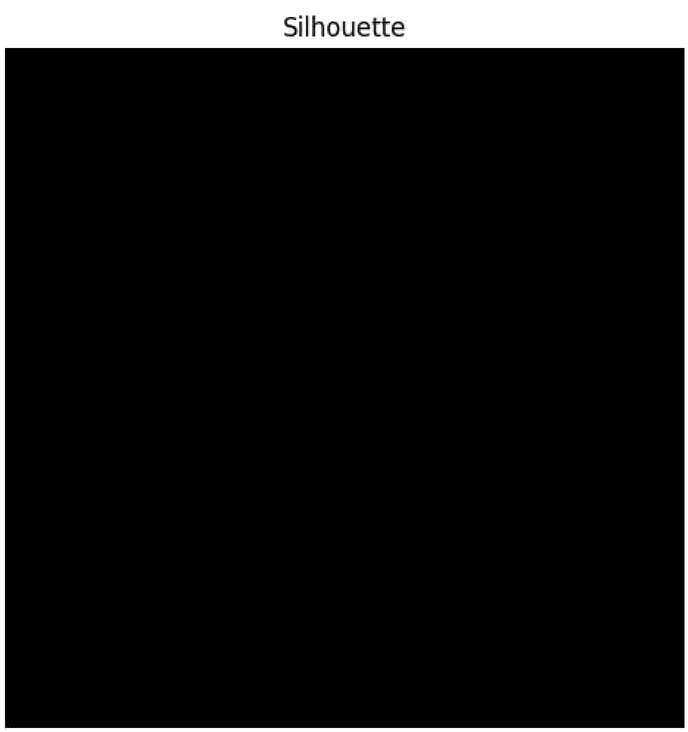

Figure 9: Silhouette - silh=0, tex=1

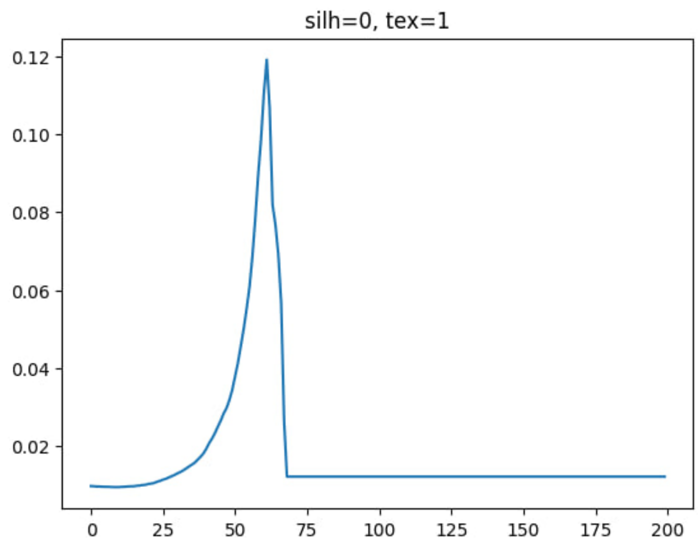

Figure 10: Loss - silh=0, tex=1

- Optimization becomes unstable
- Camera moves away from the object
- Rendered image becomes empty

Conclusion: Texture alone is insufficient for reliable pose estimation.

4.3 Combined (1, 0.1)

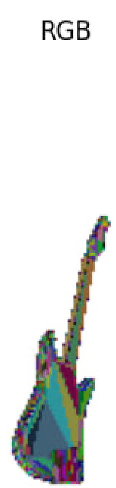

Figure 11: RGB - silh=1, tex=0.1

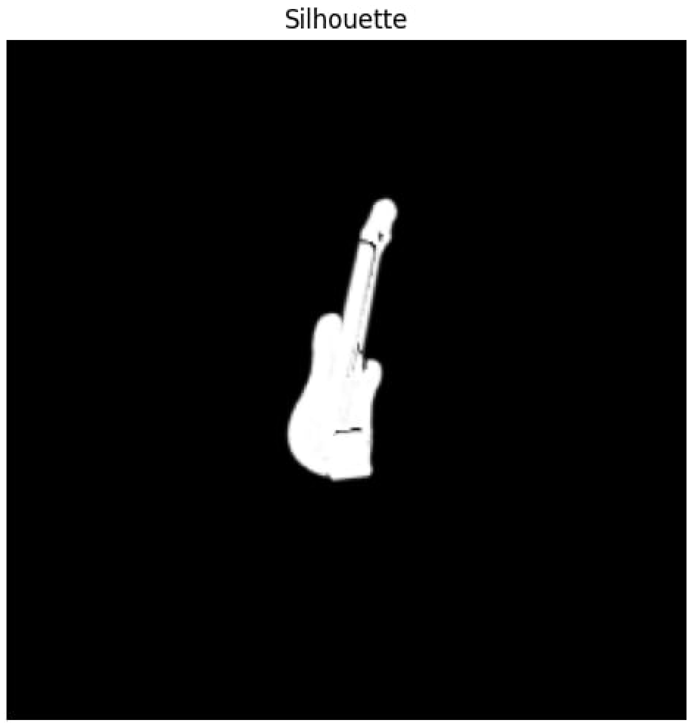

Figure 12: Silhouette - silh=1, tex=0.1

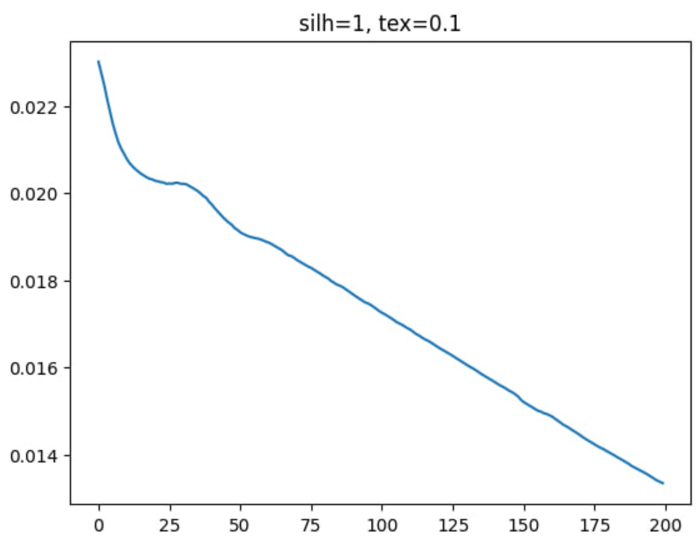

Figure 13: Loss - silh=1, tex=0.1

- Stable convergence
- Accurate pose reconstruction
- Good alignment of both shape and appearance

Conclusion: Best balance between geometry and detail.

4.4 Combined (1, 0.3)

Figure 14: RGB - silh=1, tex=0.3

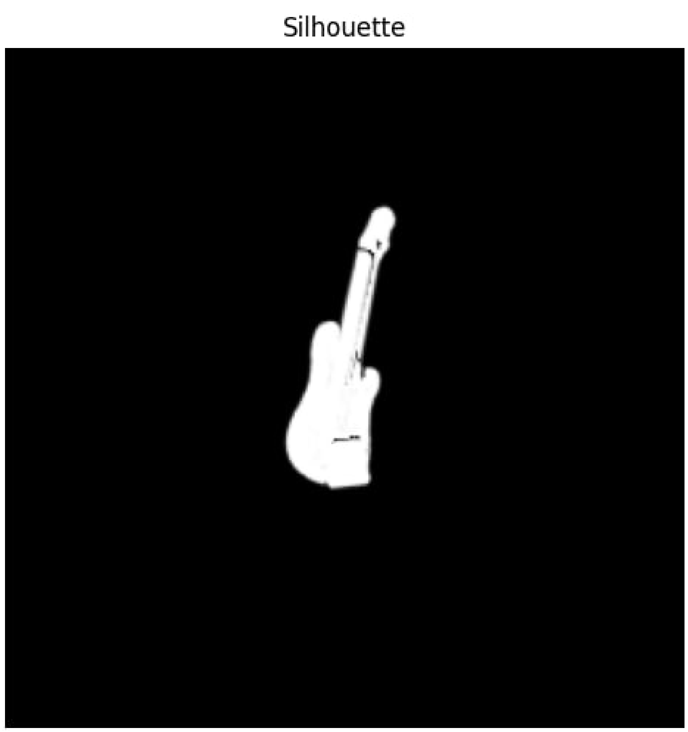

Figure 15: Silhouette - silh=1, tex=0.3

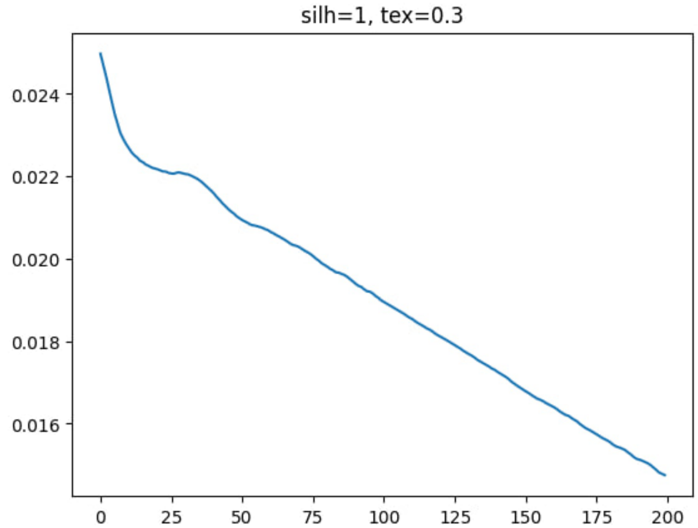

Figure 16: Loss - silh=1, tex=0.3

- Still converges
- Slightly worse than (1, 0.1)

Conclusion: Higher texture weight introduces noise into optimization.

5 Discussion

The experiments show a clear difference between silhouette and texture contributions:

- Silhouette loss
  - Provides global shape information
  - Stable and reliable
  - Guides camera optimization effectively
- Texture loss
  - Provides local detail
  - Sensitive to color variations and lighting
  - Can mislead optimization

A critical observation is that using texture loss alone may lead to degenerate solutions, where the camera moves away from the object, resulting in an empty render.

6 Conclusion

The study demonstrates that:

1. Silhouette loss is essential for accurate pose estimation.
2. Texture loss can improve results but must be used with a smaller weight.
3. The best performance is achieved with a combined loss, where silhouette dominates and texture refines the result.

Thus, the silhouette provides a robust global signal, while texture acts as an auxiliary component for fine adjustments.